Dental Significance & Challenges in Managing Patients with Traumatic Brain and Spinal Injuries

OMFS 512: Integrated Biomedical and Clinical Science

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Learning Outcomes & Competency Statements

Learning Outcomes

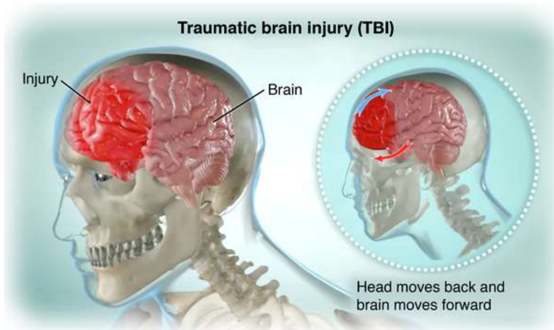
1. Review the types of traumatic brain and spinal cord injuries.
2. Discuss the dental considerations and modifications associated with treating patients with brain and spinal cord injuries.
3. Review the Americans with Disabilities Act requirements for the dental practice.

Competency statements met

1.3, 1.5, and 2.1

Traumatic Brain Injury

- Definition: Temporary or permanent damage to the brain resulting from sudden and/or violent injury.
- Affects millions of people globally each year
- Male predilection
- Accounts for a third of all injury-related deaths in the U.S.



www.saintlukeskc.org/health-library/traumatic-brain-injury

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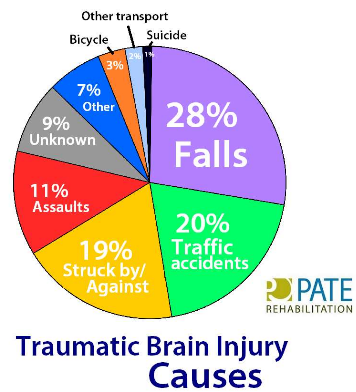
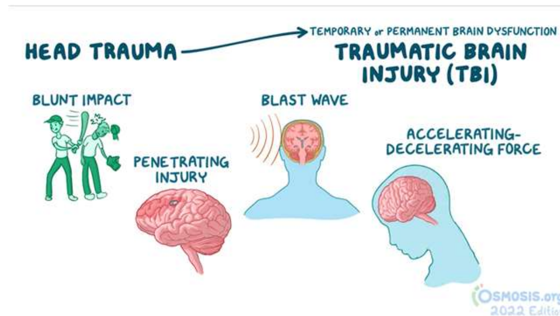
Traumatic Brain Injury (TBI)

- Causes of TBI-related deaths by age group:
  - 0-4 years: Violent assault
  - 5-24 years: Motor vehicle accidents
  - 25-64 years: Intentional self-harm
  - 65+ years: Injuries from falling



Causes of Traumatic Brain Injury

- Blunt trauma
- Penetrating wounds
- Blast waves
- Accelerating/Decelerating Forces



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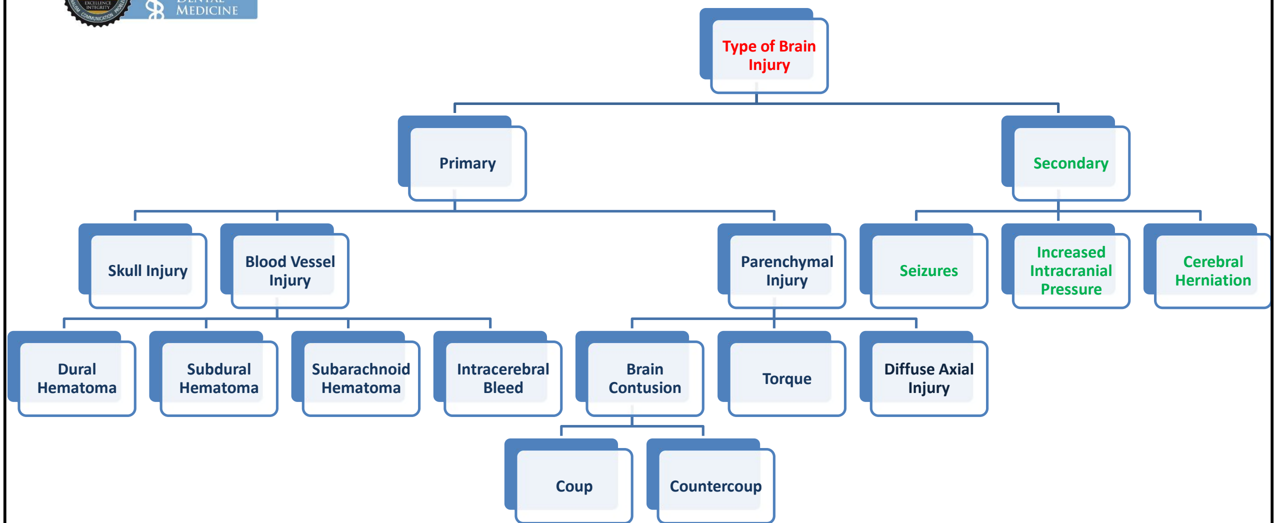
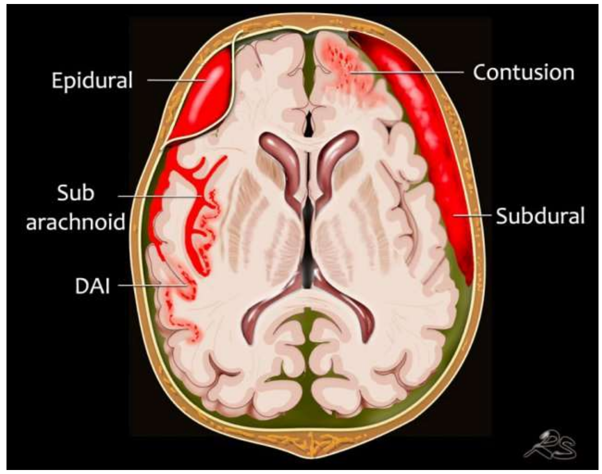

Type of Brain Injury
Primary
Secondary
Skull Injury
Blood Vessel Injury
Parenchymal Injury
Seizures
Increased Intracranial Pressure
Cerebral Herniation
Dural Hematoma
Subdural Hematoma
Subarachnoid Hematoma
Intracerebral Bleed
Brain Contusion
Torque
Diffuse Axial Injury
Coup
Countercoup

Traumatic Brain Injury

- **<u>Primary Head Injury</u>**
- **Occurs at the time of injury**
- **Focal or diffuse**
  1. **Skull fracture(s)**
  2. **Blood vessel injury**
     - Epidural hematoma
     - Subdural hematoma
     - Subarachnoid
     - Intracerebral hematoma



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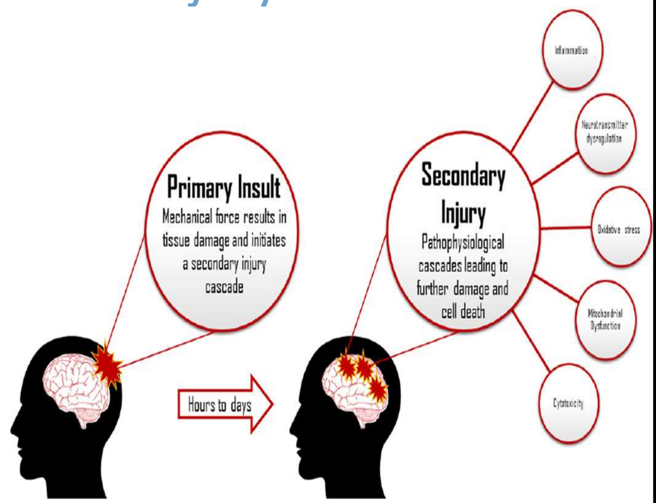
Traumatic Brain Injury

- **<u>Primary Head Injury</u>**
  - **Brain parenchymal injury**
    - Brain contusion
      - Coup - contusion at site of impact
      - Countercoup - contusion opposite from the site of impact
    - Torque – twisting around the brain stem
    - Diffuse axonal injury – shearing/tearing on the axonal portion of neurons

Traumatic Brain Injury

- <u>**Secondary Head Injury**</u>
  - **Occurs hours or days after initial injury**
    - Seizures
    - Increased intracranial pressure
    - Cerebral herniation



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ssification on Traumatic Brain Injury

- Mild
  - AKA Concussion
  - Chronic Traumatic Encephalopathy
- Moderate
- Severe

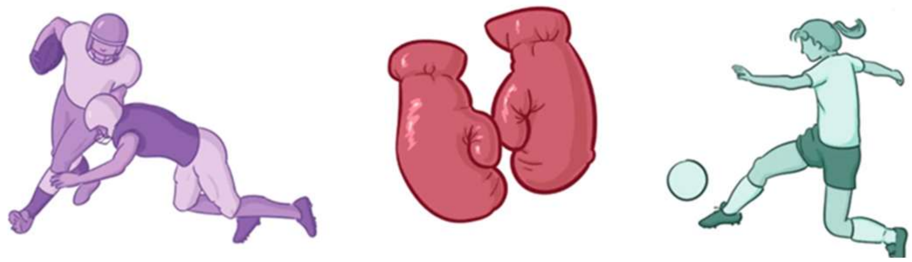
Glasgow coma scale (GCS)

Eye Opening (E)	Verbal Response (V)	Motor Response (M)
4 Spontaneously	5 Normal conversation	6 Normal
3 To voice	4 Disoriented conversation	5 Localizes to pain
2 To pain	3 Words, but not coherent	4 Withdraws to pain
1 None	2 No words, only sounds	3 Decorticate posture
	1 None	2 Decerebrate
		1 None

Total GCS = E score + V score + M score (minimum 3; maximum 15)
Score: 13–15 = Mild TBI; 9–12 = Moderate TBI; <8 = Severe TBI

CHRONIC TRAUMATIC ENCEPHALOPATHY (CTE)

* REPETITIVE HEAD TRAUMA - CONTACT SPORTS



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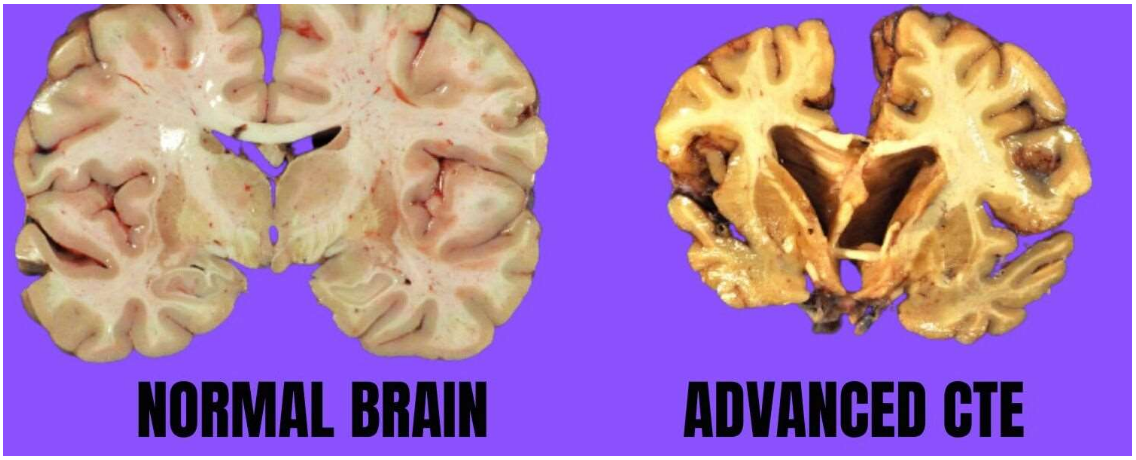
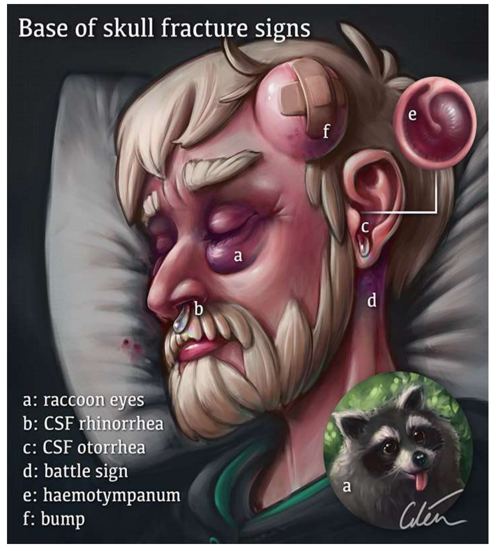
Chronic Traumatic Encephalopathy (CTE)



Traumatic Brain Injury

- Basilar Skull Fracture
  - Periorbital edema (Racoon eyes)
  - Bruise behind ear (Battle sign)
  - CSF leakage from the nose and ear
    - Rhinorrhea
    - Otorrhea



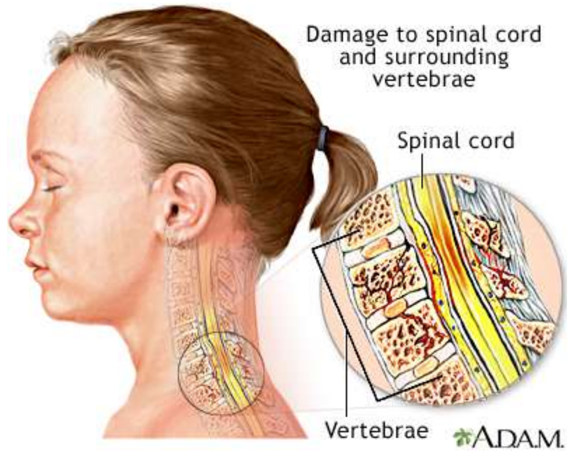
Traumatic Brain Injury

- Trauma to oral and facial structures
  - Fracturing of the maxilla and mandible
  - Dislocation of the TMJ
  - Fracturing of teeth
  - Avulsion
  - Laceration of the oral soft tissues

www.jems.com/patient-care/emergency-trauma-care/assessing-and-managing-facial-trauma/

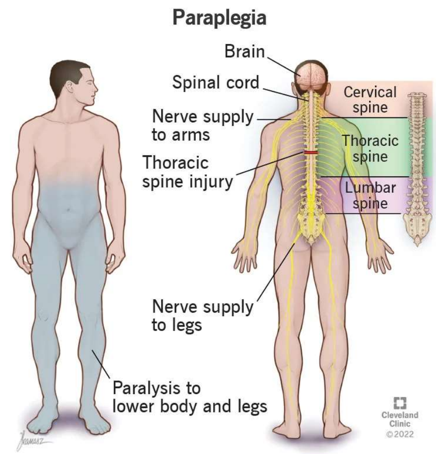
Spinal Cord Injury

- Spinal cord injury can affect sensory and motor conduction
  - **Quadriplegia**
    - Impairment or loss of motor and sensory function in the cervical segment of the spinal cord.
    - Loss of function in the trunk and upper/lower extremities



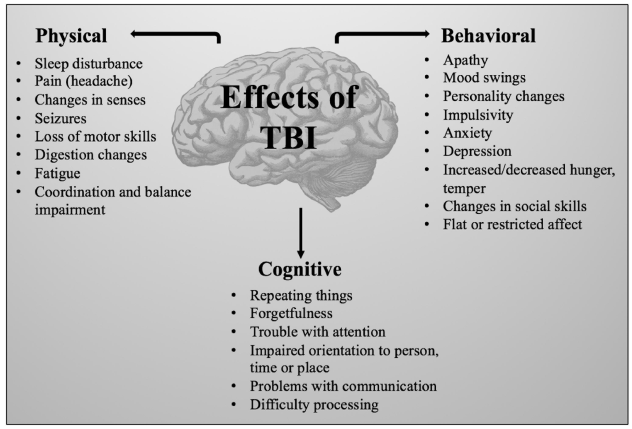
Spinal Cord Injury

- **Paraplegia**
  - Impairment or loss of motor and sensory function in the thoracic, lumbar, and sacral portions of the spinal cord.
  - Loss of function in the trunk, legs, and pelvic region.



Neurologic Disease Associated with Traumatic Brain Injury

- Stroke
- Parkinson's Disease
- Alzheimer's Disease
- Epilepsy
- Multiple Sclerosis
- Amyotrophic Lateral Sclerosis



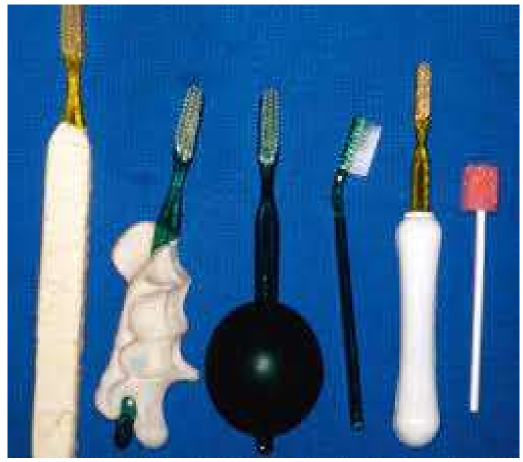
Sequelae of Traumatic Brain and Spinal Cord Injury

- Cognitive
  - Memory loss, changes in speech, and developmental delay
  - Loss of concentration and disinhibition
- Behavioral & Psychologic
  - Depression, PTSD, anxiety, and insomnia
- Physical Disabilities
  - Paralysis, hemiparesis, difficulty with processing sensory information, muscular weakness, and respiratory difficulty

Dental Considerations and Modifications

- Lack of manual dexterity
- Adverse reactions of medications
- Providing oral health education to caregivers
- Impact of oral health on the quality of life
- Access to dental services
- Dental office accommodations for patients with disabilities

Dental Considerations and Modifications



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Dental Office Accommodations for Patients with Disabilities

- Handicapped parking
- Installing an elevator or ramps
- Widening halls and doorways
- Adding handicap-accessible restrooms and drinking fountains
- Decreasing the height of curbs in certain areas
- Ensuring you have appropriate flooring surfaces
- Using proper signage for the visually impaired
- Altering counter top height
- Changing shelving or coat rack height

Resources

1. Henry RG. Chapter 14: Neurological Disorders. *The ADA Practical Guide to Patients with Medical Conditions*, Second Addition. Edited by Lauren L. Patton and Michael Glick. John Wiley & Sons. Inc. 2016, 299-324.
2. Kothari M, Pillai RS, Kothari SF, Spin-Neto R, Kumar A, Nielsen JF. Oral health status in patients with acquired brain injury: a systematic review. *Oral Surg Oral Med Oral Pathol Oral Radiol*. 2017 Feb;123(2):205-219.e7.
3. Slade GD. Derivation and validation of a short-form oral health impact profile. *Community Dent. Oral Epidermiol.* 1997;25: 284-90.
4. Takeuchi S, Planerova A, Malmstrom H, Saunders R. Improving oral health-related quality of life for a traumatic brain injury patient: A case report. *Spec Care Dentist*. 2019;39:617-623
5. University of Washington School of Dentistry. Oral Health Fact Sheet for Dental Professionals: Adults with Traumatic Brain Injury. TBI-Adult.pdf (washington.edu).
6. https://www.osmosis.org/learn/Clinical_Reasoning:_Traumatic_brain_injury
7. https://www.osmosis.org/learn/Concussion_and_traumatic_brain_injury